

Anmerkung Conditions Report

Rule engine covering Dachser, K+N, DHL Express, Wackler & Honold · source: assets/anmerkung.js · generated 2026-10-09

How a row is classified

Each worksheet row is read column-wise into the five *process** functions. A row is only classifiable when its **Stat_Freigabe** cell equals **10** (released); otherwise the row is skipped (Wackler still runs its Kontierung check). Each forwarder compares its forwarder-specific **Differenz** columns (FR, MT, SNK, EXP, TZ, ...) against a **firing threshold**:

`hasErr(v, t) = |v| > t`

A delta that clears the threshold is called a "trigger" and contributes a label. All labels a row gathers are joined with " **//** " into the **Anmerkung** cell (deduplicated by join). Some rules *return early* or block later rules, because a single finding (e.g. a duplicate Fremdnummer booking) makes every other delta on the row spurious.

Thresholds

Constant	Value
Firing threshold — Dachser	<code> diff > ?</code> (default 0.08)
Firing threshold — K+N	<code> diff > ?</code> (default 0.09)
Firing threshold — DHL Express	<code> diff > ?</code> (default 0.04)
Firing threshold — Wackler	<code> diff > ?</code> (default 0.09)
Persisted per-user	localStorage key <code>anme</code>
Kontierung? checks	disabled globally — <code>KONTIERUNG_ENABLED = fal</code>

1. Dachser — processDachser

Gate: Stat = 10. Reads C502/C503 (Einlagern/Auslagern), SNK, ZZ, SAM, DGR, EXP, MT, TZ, C38L, FR with weight/tier logic on the DACHSER_BP rate-card tiers (50 kg steps → 10 000 kg, open 999 999 bucket). Cascade order matters — the first rule that fires colours the rest.

1.0 Blank-Tarif accounting rows (early return)

Condition	Label written to Anmerkung
TARIF cell empty AND FR > T; no SACH, no SERV_ART, no other diff (MT/TZ/EXP/SNK/ZZ/DGR/LG/AV/SAM/SBFU/C38L)	VORHOLUNG
same empty-TARIF but SACH + SERV_ART are set	Fremdnummer <row's ReferenzNr> bereits berechnet in RE00123xxx, ok? (falls back to the 5034xxx placeholder when ReferenzNr is empty; the Beleg it was charged in stays a placeholder, it names a document outside this workbook)
empty TARIF + FR trigger + departure country non-DE	kein Tarif für <Land>
empty TARIF + FR trigger + domestic lane above 10 t (Brutto kg > 10000)	Kein Tarif für DE >10 to, ok?

1.1 Direct-cost labels (each fires independently)

Condition	Label written to Anmerkung
C502 Kosten DL has a non-zero value	Einlagern
C503 Kosten DL has a non-zero value	Auslagern
LG Differenz over threshold	Lagergeld
AV Differenz over threshold	Gebühr für vergeblichen Abholversuch
ZZ Differenz over threshold	2. Zustellung
SAM Differenz over threshold	Samstagszustellung
DGR Differenz over threshold (skipped when TARIF is zero)	Gefahrgut-Zuschlag
EXP Differenz over threshold	Produktzuschlag, or Terminzuschlag when EXP Kosten DL = 95
ZABF Differenz over threshold	Einfuhrzollabfertigung
SBFU Differenz over threshold	SBfU-Bescheinigung f. Umsatzsteuerzwecke
C38L Differenz over threshold	Maut von/bis NL

1.2 SNK — switch on billed SNK_DL

Condition	Label written to Anmerkung
SNK_DL = 190 or 95	AUSFALLFRACHT
SNK_DL = 130 or 390	Standgeld
SNK_DL = 75 + SNK_TARIF > 0 + SNK diff over T	Differenz Hebebuehnen-Zuschlag
SNK_DL = 75 + SERV_ART = K1AV	Speditionskosten gem. Text
SNK_DL = 75 (no tariff backing)	Ausfallfracht/Schadensersatz
SNK_DL = 80 + SERV_ART = K1AV	Speditionskosten gem. Text // Admin Zeitfensterbuchung Handel
SNK_DL = 11 (+ diff over T)	Differenz Telefonische Zustellankündigung - Laderaumzuschlag

SNK_DL = 14 (+ diff over T, TARIF not zero)	Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag (K1AV → Differenz Laderaumkostenentwicklung)
SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF=0)	Admin Zeitfensterbuchung Handel
SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF>0)	Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag
SNK_DL = 5 (+ diff over T); skipped when TARIF zero	Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag
SNK_DL = 9 (+ diff over T, TARIF not zero)	Differenz Telefonische Zustellterminvereinbarung - Laderaumzuschlag
other SNK_DL + SERV_ART = K1AV, or non-integer SNK_DL & SNK_DIFF (SNK_DIFF >= T when non-integer, else > T)	Differenz Laderaumkostenentwicklung
other SNK_DL (+ diff over T)	Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag
Non-integer SNK_DL and SNK_DIFF rounds to 5/9/11/14 (±0.05)	code re-derived from SNK_DIFF, then the switch above

1.3 EXP + others

Condition	Label written to Anmerkung
EXP diff over T and EXP_DL = 95	Terminzuschlag
EXP diff over T (otherwise)	Produktzuschlag
Empf.-Ort = LONDON, PLZ leading letter	Zone korrekt berechnet? (+ Einfuhrzollabfertigung when EXP clean)
ZABF Differenz over threshold (Import-Zollabfertigung line)	Einfuhrzollabfertigung

1.4 FR — weight/tier decision tree (after SNK)

Condition	Label written to Anmerkung
Anz. Sdg > 1 (multi-consignment)	hätte gebündelt werden können?
REFERENZ3 = ZW (deviating intermediate consignee)	Differenz aufgrund abweichender Zwischenempfänger <PLZ Ort>
SERV_ART = K1AS	Sonderfahrt
SERV_ART = K1AU and FR – 90 ≤ 0.1	"Sonderfahrt" 90 EUR doppelt berechnet?
weights equal or same tier; FR > 1 on an international lane (non-DE origin or destination)	Sonderfahrt, or Dachser berechnet die Kosten nach dem bisherigen Tarif – decided by the rate cards: (FR Kosten DL – S) landing on a NEW-card cell for S in 180/150/120/90 gives Sonderfahrt, otherwise FR Kosten lt. Tarif landing on an OLD-card cell (timed by 1.15 on imports) gives the bisherigen wording. Ratecards are encrypted (assets/dachser-ratecards.enc.json); without the passphrase the bisherigen wording stands.
weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); ZZ diff = 35	Fracht Differenz (ZZ=35)
weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); otherwise	Frachtdifferenz

weights equal or same tier; FR < -1.0	Differenz aufgrund abweichender Gewichte
both weights known, different tier, FR 0-1	Frachtdifferenz
both weights known, different tier (FR ≥ 1 or negative)	Differenz aufgrund abweichender Gewichte
no weight basis, FR negative (or < 1.0)	Differenz Frachtzu/ abschlag
no weight basis, FR positive	Differenz aufgrund von abweichendem Gewicht
Condition	Label written to Anmerkung
SNK_DL = 14 + SNK diff over T (footer)	Abholterminvereinbarung
row has no other label, only TZ over T	Differenz treibstof

2. K+N — processKN

Gate: Stat = 10. Tiers from KN_BP (like Dachser but collapses the 8500/9500 steps and opens at 99 999). FR (freight) is the lead column; EXP/MT/SNK/TZ add their own labels.

Condition	Label written to Anmerkung
FR over T and ReferenzNr lists multiple docs (contains ',') — early return	hätte gebündelt werden müssen, ok?
FR over T, TARIF blank or '-' — flat-rate freight	Pauschalfracht
FR over T, Amazon recipient, VKG and VKG_DL in the same weight tier	hätte nach Amazon Tarif abrechnen müssen
FR over T, Amazon recipient, VKG and VKG_DL cross tiers	Differenz aufgrund abweichender Gewichte
FR > 0 and SNK_DL = 5, same tier	hätte nach Amazon Tarif abrechnen dürfen
FR > 0 and SNK_DL = 5, cross tier	Differenz aufgrund abweichender Gewichte
any other FR over T	Differenz aufgrund abweichender Gewichte
EXP over T	FIXTERMIN
MT over T	Mautdifferenz

SNK cascade — exact matches first, then tolerance fallback

Condition	Label written to Anmerkung
SNK diff $\approx$ +9	Avis, ok?
SNK diff $\approx$ -9	Differenz avis
SNK diff $\approx$ 25	Portalavisierung, ok?
SNK diff over T and SNK_DL = 5 or 25	Portalavisierung, ok?
SNK diff over T and SNK_DL = 9	Avis, ok?
SNK diff over T and SNK_DL = 12	B2C-zuschlag, ok?
SNK diff over T and SNK_DL = 18 (unless Avis already)	Avis, ok?
SNK diff over T and SNK_DL = 34	Portalavisierung, ok? // Avis, ok?
SNK diff over T, code not matched	SNK Differenz
TZ over T AND neither FR nor MT fired	Differenz treibstoff
KOST or SACH blank/'-' (disabled)	Kontierung?

3. DHL Express — processDHL

Gate: Stat = 10. Uses a **blocking chain**: the first group (EL/FR/PAL/OW/YO/YL/ND/SF + SNK) sets block, and only when nothing blocked does the AC/MT/NX/OS group fire.

Condition	Label written to Anmerkung
TARIF present but value 0 (raw contains '0' or is '-')	Fremdnummer xxx bereits berechnet in RExxx, ok?.
EL Differenz over T	EXTENDED LIABILITY, ok?
FR Differenz over T	Differenz aufgrund von abweichendem Gewicht/Volumen
PAL Differenz over T	nicht stapelbar ok?
OW Differenz over T	Differenz overweight, ok?
YO positive and divisible by 15	Non conveyable piece-weight ok?
YO over T (not a 15-multiple)	non conveyable piece ok?
YL over T	Non-conveyable piece irregular ok?
ND over T	Neutral delivery ok?
SF over T	Direct signature ok?
SNK = 5	DIRECT SIGNATURE , ok?
SNK = 25	Limited quantities ok?
SNK = 30	Elevated Risk, ok?
SNK = 60	Elevated risk ok? // Restricted destination ok?
SNK over T, code not matched	SNK Differenz
below-block group — AC = 11	Addres Correction, ok?
below-block group — AC over T	Address Correction ok?
below-block group — MT over T	Mautdifferenz
below-block group — NX over T	demand surcharge ok?
below-block group — OS over T	Oversize piece ok?
no other label, TZ over T only	Differenz treibstof

4. Wackler — processWackler

Gate: Stat = 10 (otherwise only the Kontierung check runs). Fourteen numbered rules evaluated in order with several early returns. Weight-tier logic uses the Wackler rate-card breakpoints (50 kg steps → 10 000 kg, open 999 999 bucket) shared by the national and international rate sheets.

Terminal (early-return) rules

Condition	Label written to Anmerkung
TARIF '-' or '0' (non-empty), or blank + FR/MT/TZ signal — duplicate booking; all delta rules suppressed	Fremdnummer xxx bereits berechnet in RExxx, ok?
TARIF blank + SNK signal + no FR/MT/TZ + no known SNK/AVIS code — storage fee	Lagergeld, ok?
SNK $\geq 1.0 \times \text{TARIF}$ + no FR/MT/TZ + no codes (placeholder tariff)	Pauschalfracht, ok?

AVIS surcharge codes

Condition	Label written to Anmerkung
AVIS = 7.5 / 8.5 / 6.5 / 8.7 (sign-insensitive)	Avis, ok?
AVIS = -8.7 (should have used the cheap telephonic rate)	hätte Avisgebühr telefonisch abrechnen dürfen
AVIS = 1 (per-shipment Avisnachweis line)	Differenz avis, ok?

SNK surcharge code book (sign-insensitive, tolerance ± 0.5)

Condition	Label written to Anmerkung
SNK ≈ 38	NL-FIX
SNK $\approx 11.5 (\pm 0.1)$	hätte B2C-Line abrechnen dürfen
SNK ≈ 22	2. Zustellung ok?
SNK ≈ 25	Terminzustellung
SNK ≈ 80 on a weighed shipment (weight-gated)	Terminzustellung
SNK ≈ 170	Terminzustellung
SNK ≈ 180	Terminzustellung, ok?
SNK ≈ 43	2.Zustellung ok?
SNK ≈ 289	Umverfügung
SNK ≈ 49	NL-12, ok?
SNK ≈ 113	NL-SPEZ, ok?
Empf. = 88499 RIEDLINGEN	Return, ok?
SNK ≈ 16	NIGHTFIX
SNK ≈ 81	NL-10, ok?
SNK ≈ 250	"Zusatzkosten 250 €", ok?

Gewichte / bundling decision tree (FR + both weights real)

Condition	Label written to Anmerkung
multi-ref, weights 1–5% apart (same or cross tier)	hätte gebündelt werden müssen
VKG and VKG_DL in different rate tiers	Differenz aufgrund abweichender Gewichte
same tier, ≥ 6 references	hätte gebündelt werden müssen
pallet volume ($\text{colli} \times 285$) $\geq 10\,000$ kg or $\geq 1.05 \times \text{VKG}$, tiers apart	Differenz aufgrund abweichender Gewichte

same tier, ≥ 3 references (fallback)	Differenz aufgrund abweichender Gewichte
otherwise (same-tier rate-card case)	Wackler rechnet Frachtrate für <tier>kg ab
cross-tier, multi-ref, VKG_DL $\leq 20\%$ of VKG — TERMINAL	hätte gebündelt werden müssen (alone)
cross-tier, multi-ref, FR > 0 , weights within 20%	hätte gebündelt werden müssen
cross-tier, far weights or FR credit	Differenz aufgrund abweichender Gewichte
same-tier multi-ref with unequal whole-kg weights	hätte gebündelt werden müssen
same-tier multi-ref with (near-)identical weights, non-integer (volumetric) VKG	Wackler rechnet Frachtrate für <floor tier>kg ab
no real weights + multi-ref + FR	hätte gebündelt werden müssen

Fallback / additive rules

Condition	Label written to Anmerkung
FR over T, no Gewichte/bundling/return fired	Frachtdifferenz
MT over T	Mautdifferenz
$ SNK \geq 5.0$, unrecognised code, not bundled	SNK Differenz
$SNK \approx -150 (\pm 2)$	hätte Hebebühne abrechnen dürfen
$ TZ \geq 2.0$, not on same-tier 'Wackler rechnet' row, not an FR credit — on a Hebebühne credit row the fuel gate queries the surcharge instead of flagging it	Differenz Energiezuschlag // Dieselizeuschlag ok?
NL-FIX + FR + blank KOST/SACH — 'Zone korrekt?' prepended, Frachtdifferenz stripped	Zone korrekt? // ...
only rule left is TZ over T	Differenz Energiezuschlag
KOST & SACH both blank/X (disabled)	Kontierung?

Wackler constants

Constant	Value
$ TZ $ trigger (rule 11)	2.0
$ SNK $ noise floor (rule 10)	5.0
$SNK \approx -150 \pm 2 \rightarrow$ Hebebühne	150 ± 2.0
$SNK \geq \text{ratio} \times \text{TARIF} \rightarrow$ Pauschalfracht	$\times 1.0$
$SNK \approx 80 \pm 0.5 \rightarrow$ Terminzustellung (weighed)	80 ± 0.5
Multi-ref same-tier upper bundle weight	1000
Cross-tier near band (bundling)	0.2
$VKG_DL \leq$ fraction of VKG $\rightarrow$ terminal bundling	0.2
Pallet volume estimate per collium (kg)	285
'Near-equal' weights band	0.01
Close-but-unequal bundle band	0.05
Same-tier $\geq N$ references $\rightarrow$ bundling	6

5. Honold — processHonold

Gate: Stat = 10, plus the OLD tariff must be unlocked. The tariff matrix is business data and ships encrypted (assets/honold-tariff.enc.json); until it is decrypted the engine *skips* Honold rows (null) rather than blanking an existing note it cannot re-derive. One rule:

Condition	Label written to Anmerkung
FR Kosten DL is looked up in the destination country column of the OLD tariff to find the weight bracket(s) whose rate equals it; Volumen kg sits in one of those brackets	Honold berechnet die Kosten nach dem bisherigen Tarif
otherwise (blank / 0 FR, missing / 0 Volumen kg, unknown country, or FR matching no rate)	Differenz aufgrund abweichender Gewichte

Honold constants

Constant	Value
FR ↔ OLD-tariff match tolerance	0.005
Weight brackets	Bis column of the OLD tariff (50 ... 10 000, last row

How the output is consumed

The **Anmerkung** cell is a ' // '-joined list of independent rule outputs. The diff tooling (classifyDiff) compares the engine output against the auditor's ground-truth column after normalising phrase spelling/case/whitespace (normPhrase), and buckets each row as **correct / missed / overfired / wrong**. Phrase-level decomposition then reports which specific phrase the engine missed (*missing_phrases*) and which it over-fired (*extra_phrases*). That comes right from the A/B training pairs — a rule change is only accepted when it re-validates against the git-ignored Soll-Ist workbooks.

Control surface: scripts/anmerkung/verify.mjs smoke-tests that the five process* functions and window.AnmerkungV5 exist; the same periods feed the diagnostics facade. Firing thresholds and the Wackler constants above are read live out of assets/anmerkung.js at build time, so this report tracks the engine — but the cascade descriptions are maintained by hand in scripts/anmerkung/report_conditions.py and should be kept in sync when rules change.